MIL-B-12504E(MR)
10 February, 1987
SUPERSEDING
MIL-S-12504D(MR)
24 October 1975

MILITARY SPECIFICATION

BAR, METAL & WIRE (NONELECTRICAL), CARBON AND ALLOY STEEL, COLD DRAWN FOR AP BULLET CORES AND SHOT

This specification is approved for use by the U.S. Army Materials Technology Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three compositions of cold drawn carbon and alloy steel bars for use in the manufacture of armor-piercing cores and shot.

1.2 Classification.

1.2.1 Chemical composition. The bars are classified by chemical composition as follows (see table I and 6.2):

G10700
Manganese-Molybdenum
Manganese-Chromium-Molybdenum

1.2.2 Condition. The bars are furnished in the following conditions as specified (see 6.2):

Lamellar pearlite annealed and cold drawn
Spheroidize annealed and cold drawn

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MSE, Watertown, MA 02172-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A /FSC 9510/

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-48 - Tolerances for Steel Wrought Products and for Centrifugally Cast Steel
FED-STD-66 - Steel. Chemical Composition and Hardenability

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage
MIL-STD-1459 - Macrograph Standards for Steel Bars, Billets, and Blooms for Ammunition Components
MIL-STD-1949 - Inspection, Magnetic Particle

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A255 - End-Quench Test for Hardenability of Steel.
A751 - Methods, Practices, and Definitions for Chemical Analysis of Steel Products
D3951 - Standard Practice for Commercial Packing.
E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Meterals.
E45 - Determining the Inclusion Content of Steel.
E112 - Determining Average Grain Size.
E381 - Method of Macroetch Testing, Inspection and Rating Steel Products, Comprising Bars, Billets, Blooms, and Forgings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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(Technical society and technical associations specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the steel bars shall be as specified in table I. A heat analysis shall be furnished with each heat of steel.

TABLE I. Chemical composition, heat (percent maximum unless range is indicated).

Element	G10700	Manganese-Molybdenum	Manganese-Chromium-Molybdenum
Carbon	0.65-0.75	0.65-0.77	0.52-0.62
Manganese	0.60-0.90	0.75-1.05	0.75-1.05
Phosphorous	0.04	0.04	0.04
Sulfur	0.05	0.04	0.04
Silicon	0.15-0.30	0.20-0.35	0.20-0.35
Chromium	-	-	0.80-1.15
Molybdenum	-	0.90-1.10	0.15-0.25
Copper	0.35	-	-

3.1.1 Product analysis. The chemical composition, as determined by product analysis, shall meet the requirements for the specified composition provided that, in a heat, the individual tolerances do not vary either above or below the ranges shown for product analysis tolerances in FED-STD-66.

3.2 Hardness. The HRC of the bars shall not exceed the values shown in table II.

TABLE II. Hardness requirements.

Condition	Rockwell C
Lamellar pearlite annealed and cold drawn	26
Spheroidize annealed and cold drawn	24

3.3 Hardenability. The hardenability value for each heat of alloy steel shall meet the minimum of HRC 60 at the 0.375 inches (9.525 mm) position of the test specimen. The carbon steel shall not be subject to the hardenability

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requirement, unless otherwise specified in the invitation for bids, contract, or order (see 6.2).

3.4 Austenite grain size. The bars shall be produced from fully killed steel with at least 70 percent of the austenite grains No. 5 or finer.

3.5 Soundness. The quality and cleanliness of the steel, determined from the macroetch test on the billet, shall be equal to or better than grades A4-B3-C4 of MIL-STD-1459. Defects D1 to D8 inclusive are unacceptable.

3.6 Decarburization. Depth of decarburization, total and partial, shall not exceed the values shown in table III

TABLE III. Decarburization values.

Nominal diameter			Decarburization, maximum	
inches	(mm)		inches	(mm)
0.2656	(6.747)		0.010	(0.254)
0.4531	(11.510)		0.012	(0.305)
0.8438-0.9063	(21.431)	(23.019)	0.017	(0.432)

3.7 Magnetic particle inspection. Bars shall be inspected by magnetic particle and show no cracks or seams.

3.8 Condition (see 1.2.2). Bars ordered to lamellar pearlite annealed and cold drawn shall show at least 50 percent lamellar pearlite, and bars ordered to spheroidize annealed and cold drawn condition shall show at least 75 percent spheroidization.

3.9 Nonmetallic inclusions. Bars shall conform to the specified nonmetallic inclusion classification of ASTM E 45 (see 6.2).

3.10 Dimensions.

3.10.1 Diameter. Unless otherwise specified in the contract or order (see 6.2), bars shall conform to the dimensions ordered within the tolerances shown in table IV.

TABLE IV. Permissible variations in diameter.

Ordered diameter		Permissible variation, minus	
inches	(mm)	inches	(mm)
0.2656	(6.747)	0.005	(0.127)
.4531	(11.510)	.005	(0.127)
.8438	(21.431)	.006	(0.152)
.9063	(23.019)	.006	(0.152)

3.10.2 Out-of-round. Bars shall not be out-of round by more than one half the permissible variation in diameter shown in table IV.

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3.10.3 Length. Bars may be ordered in random, exact, or multiple lengths as specified (see 6.2). Tolerance in length shall be in accordance with FED-STD-48.

3.10.3.1 Random lengths. Bars specified to random lenghts shall be furnished in lenghts of 6 to 20 feet (1.829 to 6.096 meters), provided that not more than 20 percent by weight of a lot shall be in lengths less than 10 feet (3.048 meters).

3.10.3.2 Exact or multiple lengths. Bars specified to exact or multiple lengths shall be furnished to the specified length within the permissible variation except that bars shorter than the specified length but not less than 6 feet long (1.829 meters) may be furnished up to 10 percent of the weight of the lot. All short lengths shall be packed in bundles of equal length bars.

3.11 Identification marking. Bars shall be marked for identification in accordance with the contract or order (see 6.2).

3.12 Finish. Bars shall be furnished with an as-cold-drawn finish.

3.13 Workmanship. Bars shall have a smooth surface and be of uniform quality free from harmful defects such as pipe, porosity, and other defects which might be detrimental to the material for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the invitation for bids, contract, or purchase order (see 6.2), a lot shall consist of all bars of the same heat, of the same diameter, and same annealing charge. For continuous type annealing the lot size shall be 25 tons or less of the same size bar of each heat and annealed in the same cycle.

4.3 Sampling.

4.3.1 Chemical analysis. At least one sample shall be taken from each lot for product analysis.

4.3.2 Hardness. Samples shall represent not less than 2 percent or a maximum of 25 bars, of each lot.

4.3.3 Hardenability. At least one sample shall represent each heat of steel. When specified (see 6.2), the samples shall be heat treated to a lamellar pearlite or spheroidize condition prior to quenching for the hardenability test.

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4.3.4 Austenite grain size. At least one sample shall represent each heat for determination of austenite grain size.

4.3.5 Soundness. Samples shall be taken from the top end of the top billet and the bottom end of the bottom billet cut from the first and last market ingot. Steel that can be identified as having been previously tested and approved will be satisfactory for applications or orders to this specification. If it is desired to apply miscellaneous blooms, billets, or bars from heats not previously tested in the required manner, both ends of each bloom, billet, or bar shall be sampled for soundness testing.

4.3.6 Conditions. Two samples shall represent each lot for metallographic examination.

4.3.7 Decarburization. Two samples shall represent each lot.

4.3.8 Nonmetallic inclusions. Unless otherwise specified (see 6.2), two specimens shall be taken from each lot for nonmetallic inclusion determination.

4.4 Examination.

4.4.1 Visual. A representative number of bars shall be examined for compliance with the identification marking (see 3.11), finish (see 3.12), and workmanship (see 3.13) requirements.

4.4.2 Dimensional. A representative number of bars shall be measured for compliance with the requirements for dimensions (see 3.10).

4.4.3 Preparation for delivery. Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.5 Test.

4.5.1 Chemical composition. Chemical analysis shall be conducted in accordance with the applicable methods specified in ASTM A751.

4.5.2 Hardness. Hardness test shall be conducted in accordance with ASTM E18.

4.5.3 Hardenability. Hardenability tests shall be conducted in accordance with ASTM A255.

4.5.4 Austenite grain size. Austenite grain size determination shall be conducted in accordance with ASTM E112.

4.5.5 Soundness. Macro-etch test for soundness shall be conducted in accordance with ASTM E381. Specimens shall be compared to the standards in MIL-STD-1459 for conformance to the requirements for soundness (see 3.5).

4.5.6 Condition. The specimens shall represent the entire cross section of the sample bars, and shall be polished, etched, and examined metallographically to determine conformance with the specified condition (see 1.2.2 and 3.8).

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4.5.7 Decarburization. The specimens shall represent the entire cross section of the sample bars, and shall be polished, suitably etched, and examined at 100X magnification to determine the depth of decarburization.

4.5.8 Magnetic particle inspection. All bars shall be inspected by magnetic particle for compliance with 3.7. Inspection shall be in accordance with MIL-STD-1949.

4.6 Rejection and retest.

4.6.1 Rejection. Unless otherwise specified (see 6.2) where one or more test specimens fail to meet the requirements of this specification the lot represented by the specimen or specimens shall be subject to rejection.

4.6.2 Retest. When no sampling plan is provided or approved by the procuring agency (see 6.2) and where there is evidence that indicates that the specimen was not representative of the lot of material, and when the detail specification does not otherwise specify, at least two specimens shall be selected to replace each test specimen which failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

5. PACKAGING

5.1 Preservation. Bars shall be prepared for shipment in accordance with level A or Commercial as specified (see 6.2).

5.1.1 Level A. Cleaning, drying, preservation, and packaging shall be in accordance with MIL-STD-163.

5.1.2 Commercial. Cleaning, drying, preservation, and packaging shall be in accordance with the requirements of ASTM D3951.

5.2 Packing. Bars shall be packed for shipment in accordance with level A or Commercial, as specified (see 6.2).

5.2.1 Level A. Bars shall be packed in accordance with the requirements of MIL-STD-163.

5.2.2 Commercial. Bars shall be packed in accordance with ASTM D3951 to insure carrier acceptance and safe delivery at destination.

5.3 Marking. Marking shall be in accordance with the requirements of MIL-STD-129 or ASTM D3951, as applicable.

6. NOTES

6.1 Intended use.

6.1.1 Carbon steel. Carbon steel bars are intended for the manufacture of 7.62mm and caliber .30 AP type bullet cores.

6.1.2 Manganese-molybdenum steel. Manganese molybdenum steel bars are intended for the manufacture of caliber .50 AP type bullet cores.

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6.1.3 Manganese-chromium-molybdenum steel. Manganese-chromium-molydbdenum steel bars are intended for the manufacture of 20mm AP shot.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Chemical composition (see 1.2.1).
(c) Condition (see 1.2.2).
(d) When carbon steel is subject to the hardenability requirement (see 3.3)
(e) Nonmetallic inclusion classification (see 3.9).
(f) Diameter (see 3.10.1).
(g) Length, random, exact or multiples (see 3.10.3).
(h) Identification marking (see 3.11).
(i) When a lot size other than as specified in 4.2 is required.
(j) When the heat treatment specified in 4.3.3 is required.
(k) When other than two specimens for nonmetallic inclusion determination is required (see 4.3.8).
(l) When rejection and retest procedures other than as specified in 4.6 are required.
(m) Level of preservation and packing (see section 5).

6.3 Subject term (key word) listing.

bar, carbon steel
bullet, cores
cold drawn
killed steel
manganese - chromium - molybdenum steel
manganese - molybdenum steel

Custodian:
Army - MR

Review activities:
Army - AR
DLA - IS

Preparing activity:
Army - MR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-B-12504E(MR)	BAR, METAL AND WIRE (NONELECTRICAL), CARBON AND ALLOY STEEL, COLD DRAWN, FOR AP BULLET CORES, AND ...

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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